1 October 19th: Cyclic Groups, Subgroups

1.1 Cyclic Groups

Definition 1.1.1

Let G be a group and $g \in G$. We call the set $\langle g \rangle := \{g^n : n \in \mathbb{Z}\}$ the **subgroup of** G **generated by** g. Indeed $\langle g \rangle$ is easily verified to be a subgroup of G because $g^0 = e_G$, $g^n \cdot g^m = g^{n+m}$ and $(g^n)^{-1} = g^{-n}$. The order of this group is called the **order** of g and denoted $|\langle g \rangle|$.

Lemma 1.1.2

Let G be a group and $g \in G$ such that $|g| < \infty$. Then $|\langle g \rangle|$ is equal to the least $k \in \mathbb{N}^+$ such that $g^k = e_G$. Moreover, no such k exists if $|g| = \infty$.

Proof. Assume there exists a least $k \in \mathbb{N}^+$, then it is clear that $\langle g \rangle := \{\dots g^{-2}, g^{-1}, e_G, g, g^2, \dots\} = \{e_G, g, g^2, \dots, g^{k-1}\}$, since $g^k = e_G$ and similarly, $g^{-k} = (g^k)^{-1} = e_G^{-1} = e_G$. Thus, for any $n \in \mathbb{Z}$, there exists $q \in \mathbb{Z}$ such that

$$g^n = g^{qk+\mathrm{rem}(n,k)} = (g^k)^q \cdot g^{\mathrm{rem}(n,k)} = g^{\mathrm{rem}(n,k)}.$$

It follows that $|g| = |\langle g \rangle| \leq k$. To see that $|g| = k$, suppose by contradiction that $|g| < k$, then by the pigeonhole principle, there are two elements $0 \leq i < j < k - 1$ such that $g^i = g^j$, hence $g^{j-i} = e_G$ and $j - i \geq 1$ and $j - i < k$, contradicting, k being the least such power.

It remains to show that no such k exists if $|g| = \infty$. Indeed, if $|g| < \infty$, then $g^m = e_G$ for some $m \in [1, |g|]$, so if no such m exists, $|g| = \infty$. ∎

Definition 1.1.3

A group G is called **cyclic** if $G = \langle g \rangle$ for some $g \in G$.

Example 1.1.4. $(\mathbb{Z}, +)$ is a cyclic group with 1 being a generator, as well as -1, since every integer is a multiple of 1 or of -1.

Example 1.1.5. $(\mathbb{Z}/n\mathbb{Z}, +)$ is a cyclic group of order n, where $\overline{1}$ is a generator. In fact, if $\gcd(k, n) = 1$, then $\overline{k}$ is also a generator.

Remark 1.1.6. We will show in the future that $\mathbb{Z}$, and $\mathbb{Z}/n\mathbb{Z}$ are the only cyclic groups up to an "isomorphism".

1.2 Subgroups Generated by a Subset

It is a fact (proven in the homework) that an intersection of subgroups is a subgroup itself. The proof also works for any arbitrary intersection of subgroups. The following remark follows from this fact:

Remark 1.2.7. Let G be a group and $S \subseteq G$, (S can be empty). There exists a smallest (with respect to inclusion) subgroup $H \leq G$ such that $H \supseteq S$ i.e. $H \supseteq S$ and for every $K \leq G$ and $K \supseteq S$, $H \subseteq K$. Namely,

$$H := \bigcap_{S \subseteq K \leq G} K.$$

Definition 1.2.8

Let G be a group and let $S \subseteq G$ (S can be empty). The **subgroup generated by** S, denoted $\langle S \rangle$ or $\langle S \rangle_G$, is the least subgroup containing S, i.e. the intersection of all subgroups containing S.

Example 1.2.9. For a group G, $g \in G$, the subgroup generated by g is the same as the subgroup generated by $\{g\}$.

Example 1.2.10. In $(\mathbb{Z}, +)$, we have that $\langle \{a, b\} \rangle = \gcd(a, b)\mathbb{Z}$ because by Theorem 6.1.2, $\gcd(a, b)$ can be written as a linear combination of a and b.

1.3 Permutation Groups

Recall that for a set X, we defined the group S_X to be all bijections from X to X. These bijections are called *permutations* of X and S_X is called the *group of permutations* or *permutation group* on X. In this case, we will mainly focus on permutation groups where X is finite. For these finite X, we define $S_n := S_X$ and call S_n the *permutation group of n elements*. We denote the permutations of S_n as $\sigma(1), \sigma(2), \ldots$. If $\sigma(i) = i$, we may omit it from the notation. A permutation of n elements can be written like so:

$$\begin{pmatrix} 1 & 2 & 3 & \ldots & n \\ \sigma(1) & \sigma(2) & \sigma(3) & \ldots & \sigma(n) \end{pmatrix}$$

where $1 \mapsto \sigma(1)$, $2 \mapsto \sigma(2)$ and so on. Note that since σ is a bijection, every element that appears in the top row will appear in the bottom row.

Example 1.3.11. Consider this composition in S_4:

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 3 & 4 \end{pmatrix}$$

To compute this permutation, we apply the innermost permutation, followed by the outermost one. Let $\sigma \in S_4$ be the outermost permutation and $\tau \in S_4$ be the innermost permutation. We see $\tau(1) = 2$, so $(\sigma \circ \tau)(1) = \sigma(\tau(1)) = \sigma(2) = 4$, so $1 \mapsto 4$. Similarly, $(\sigma \circ \tau)(2) = \sigma(\tau(2)) = \sigma(1) = 2$, so $2 \mapsto 2$, and so on. The composition $\sigma \circ \tau$ is thus:

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 3 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix}$$